

SLD4B-1310-1

OVERVIEW

Superluminescent diode coupled to an optical fiber and packaged into a hermetic case

MAIN FEATURES

- Wavelength: 1310 nm
- Optical power in CW mode in single-mode fiber: 1 mW
- Package types: coaxial, coaxial with bracket, 14 pins DIL, 14 pins BTF
- Built-in monitor photodiode

ORDERING INFORMATION

SLD4B-1310-1-X-X-X-X-X-X

Case type

COAX: compact coaxial (low duty cycle pulse mode only)

COAXB: compact coaxial with a bracket

TH: compact coaxial with a bracket compatible to Thorlabs mount

DIL: common 14-pins DIL for active thermal stabilization (TEC and thermistor)

DILRAD: 14-pins DIL for active thermal stabilization (TEC and thermistor) with wall radiator

BTF: 14-pins BTF type 1 (Pump) for active thermal stabilization (TEC and thermistor)

Pinout code

9: see more details on page 5

Fiber type

SM1: SM, G.657.A1, [Corning SMF-28 Ultra](#), furcation tubing Ø0.9 mm or **BSM1** Ø0.25mm

SM3: SM, G.657.B3, [Corning ClearCurve ZBL](#), furcation tubing Ø0.9 mm or **BSM3** Ø0.25mm

SMP13: PM, [Corning PM1300](#), PANDA type, furcation tubing Ø0.9 mm

Other type on request

Connector type

FU: FC/UPC (SM1, SM3)

FA: FC/APC (SM1, SM3, SMP13)

SU: SC/UPC (SM1)

SA: SC/APC (SM1)

N: no connector (scissors cut)

Other type: on request

Test measurements

CW: CW mode (electro-optical parameters at T=25+/-5 C and spectrum)

Fiber length

0.5: 500+/-50 mm

1.0: 1000+/-100 mm

Other length on request

SLD4B-1310-1

ABSOLUTE MAXIMUM RATINGS

Parameter		Value	Unit	Conditions
SLD CW forward current	I _{max}	140	mA	CW, T = 25°C
SLD reverse voltage	V _{RL}	2	V	
Photodiode reverse voltage	V _{RP}	20	V	
Operating temperature*	T _{op}	0 - +65	°C	Coaxial package
Operating temperature*	T _{op}	-40 - +60	°C	DIL, BTF (T _{st} = 25°C)
Storage temperature	T _{stg}	-40 - +85	°C	
Soldering temperature	T _{sold}	260	°C	Max. 5 seconds

*Operating temperature is defined by the case temperature. It is necessary to ensure sufficient heat dissipation so that the module's maximum operating temperature is not exceeded. Operation at elevated temperatures reduces the lifetime of the SLD.

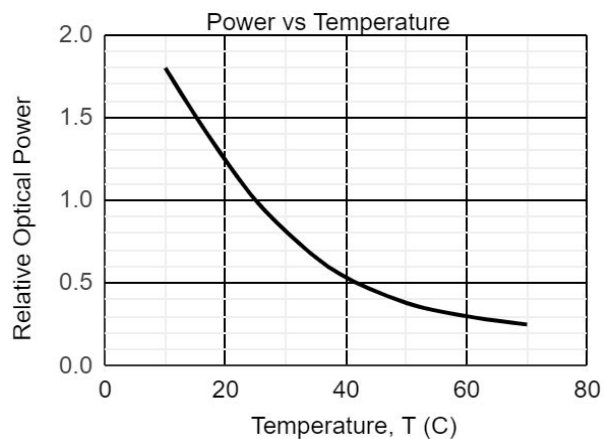
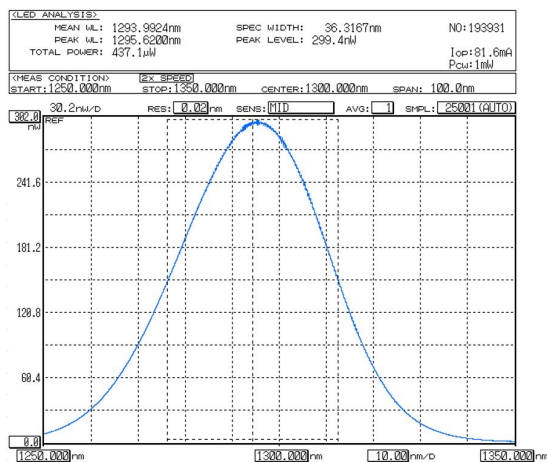
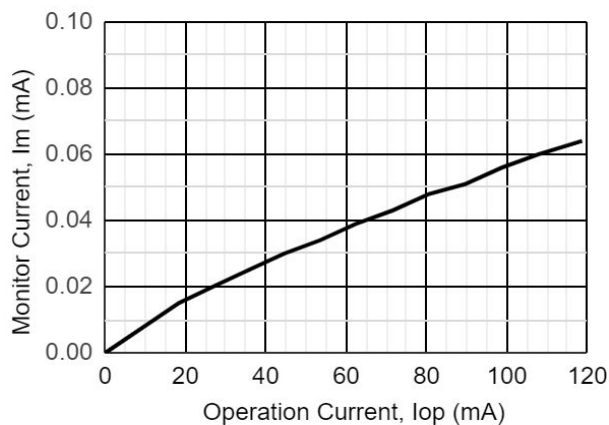
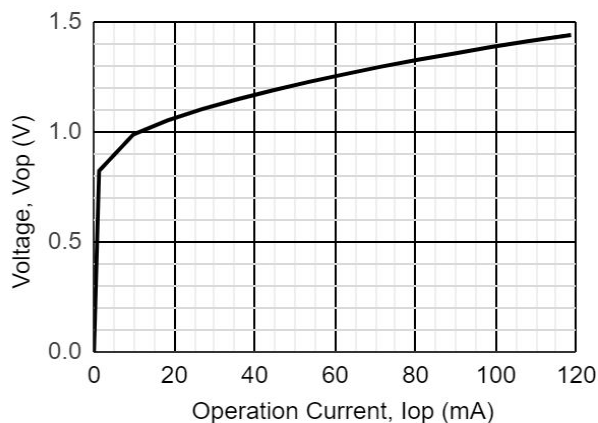
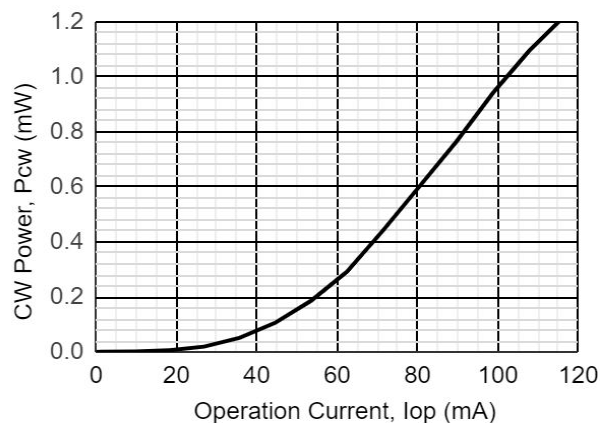
Operating temperature for the DIL, DILRAD and BTF 14-pins case with TEC is defined for internal temperature stabilization at T_{st} = 25°C that corresponds to thermistor resistance R_t = 10 kOhm.

SLD4B-1310-1

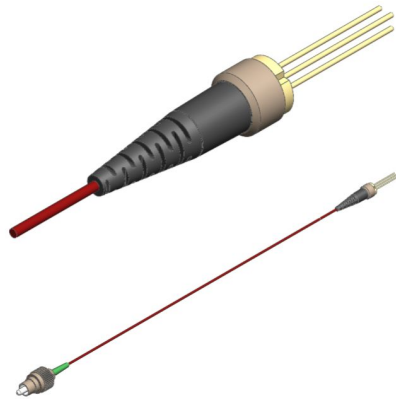
ELECTRICAL-OPTICAL CHARACTERISTICS (T = 25 °C)

Parameter		MIN	TYP	MAX	Unit	Conditions
Optical power (CW)	P _{cw}	0.8	1.0		mW	CW, I _{op} = 120 mA, SM1
Mean wavelength	λ	1280	1310	1340	nm	CW, I _{op} = 120 mA
Wavelength-temperature coefficient	dλ/dT		0.6		nm/°C	CW, I _{op} = 120 mA
Spectral width, OSA	Δλ	35	40		nm	CW, I _{op} = 120 mA, OSA, FWHM
Spectral modulation (ripple)			1	4	%	CW, I _{op} = 120 mA, BW = 1 nm
Operating voltage	V _{op}		1.4	2.0	V	CW, I _{op} = 120 mA
Cut-off frequency at -3 dB	f _c	150			MHz	I _{op} = 100 mA +/-20 mA _{p-p}
Monitor current	I _m	0.01	0.1	0.5	mA	CW, I _{op} = 120 mA, V _r = 5 V
Polarization extinction ratio	PER	17			dB	CW, SMP13

SLD4B-1310-1

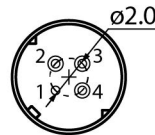


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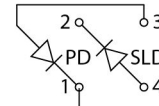


COAX

BACK VIEW



PINOUT #9



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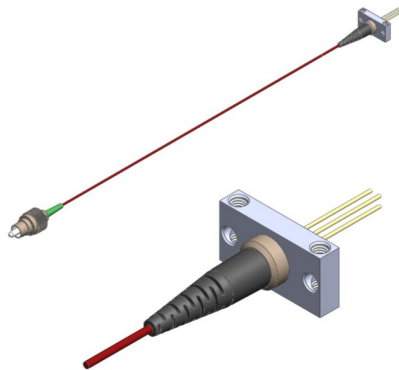
Drawing



3D model

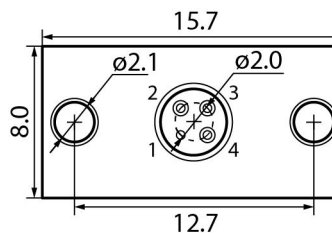


Application Notes

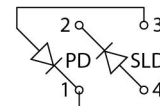


COAXB

BACK VIEW



PINOUT #9



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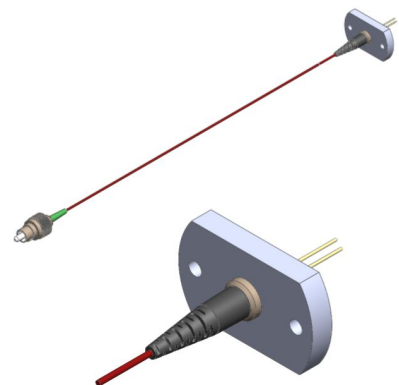
Drawing



3D model

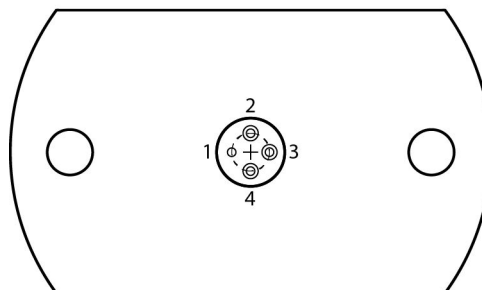


Application Notes



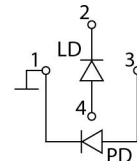
PACKAGE TH

BACK VIEW



PINOUT #9

Thorlabs Pin Code G
mPD not used



Compatible to Thorlabs
LDM9LP mount

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Drawing

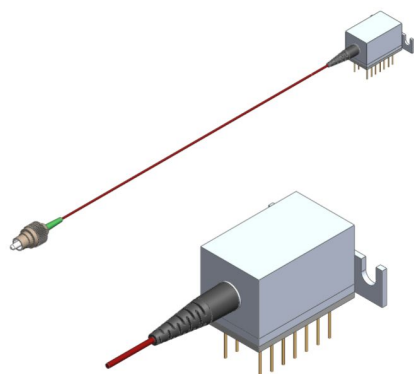


3D model

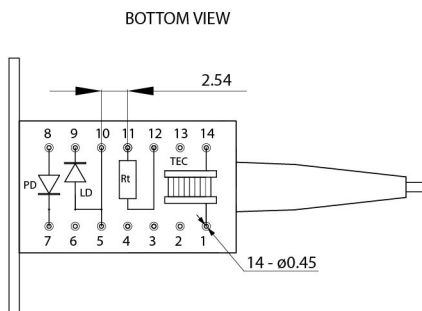


Application Notes

SLD4B-1310-1



DIL



PINOUT #1, #9, #12, #14

- 1.TEC Anode
- 2.-
- 3.-
- 4.-
- 5.LD Anode
- 6.-
- 7.PD Cathode
- 8.PD Anode
- 9.LD Cathode
- 10.LD Anode
- 11.Thermistor
- 12.Thermistor
- 13.-
- 14.TEC Cathode

TEC LP: $I_{max} = 0.7A$, $U_{max} = 3.9V$, $Q_{max} = 1.4W$,
AC R = 4.7 Ohm, $\Delta T_{max} = 72 K$

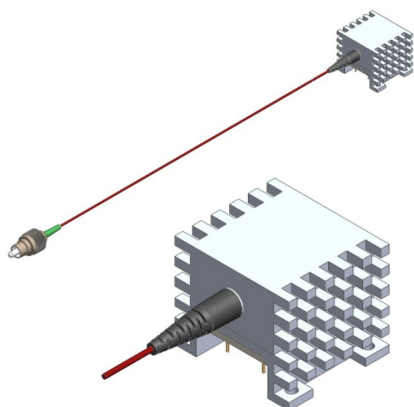
Thermistor:
 $R_t = 10 * \exp(3600 * (1/T[K] - 1/298))$ kOhm

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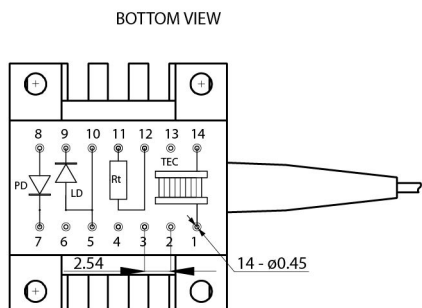
Drawing

3D model

Application Notes



DILRAD



PINOUT #1, #9, #12, #14

- 1.TEC Anode
- 2.-
- 3.-
- 4.-
- 5.LD Anode
- 6.-
- 7.PD Cathode
- 8.PD Anode
- 9.LD Cathode
- 10.LD Anode
- 11.Thermistor
- 12.Thermistor
- 13.-
- 14.TEC Cathode

TEC LP: $I_{max} = 0.7A$, $U_{max} = 3.9V$, $Q_{max} = 1.4W$,
AC R = 4.7 Ohm, $\Delta T_{max} = 72 K$

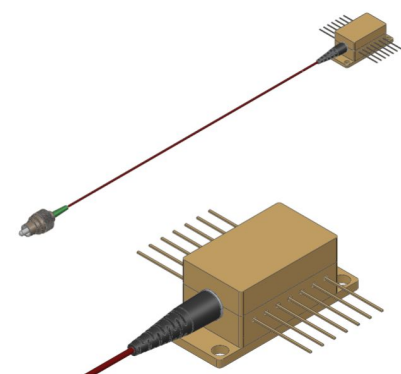
Thermistor:
 $R_t = 10 * \exp(3600 * (1/T[K] - 1/298))$ kOhm

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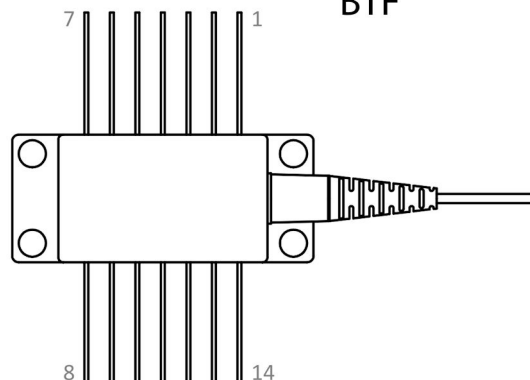
Drawing

3D model

Application Notes



BTF



1. TEC+
2. Thermistor
3. NC
4. NC
5. Thermistor
6. NC
7. NC
8. NC
9. NC
- 10 LD/SLD Anode
11. LD/SLD Cathode
12. NC
13. Case Ground
14. TEC -

TEC LP: $I_{max} = 0.7A$, $U_{max} = 3.9V$, $Q_{max} = 1.4W$,
AC R = 4.7 Ohm, $\Delta T_{max} = 72 K$

Thermistor:
 $R_t = 10 * \exp(3600 * (1/T[K] - 1/298))$ kOhm

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Drawing

3D model

Application Notes

SLD4B-1310-1

Characteristics, data, materials and structures specified in this datasheet are subject to change without notice. Please refer to the latest specification before use of the products.

Safety and handling cautions

1. Avoid smashing and burning of the module. Avoid storing and using the module in conditions where water, organic solvents or aggressive acids or bases may contact the module or where there is a possibility of exposure to corrosive gases, explosive gases, dust, salinity or other harsh conditions. The module should be disposed as special industrial waste.
2. Exceeding absolute maximum ratings even for a short time can cause permanent damage of the module.
3. The module is sensitive to and can be broken by ESD (static electricity).

Conflict Minerals Policy Statement

LD4B achieves business objectives and customer needs with social responsibility. We do not support or contribute to the violence and human rights violations associated with the mining of conflict minerals coming from Conflict Regions according to US "Dodd-Frank Act". When possible, our suppliers' conflict mineral statements are reviewed. We do not directly purchase Conflict Minerals from any source and do not knowingly procure any parts and products containing Conflict Minerals from Conflict Regions.

RoHS Compliance Statement

Restriction of Hazardous Substances (RoHS) directive (Directive 2011/65/EC amended with Directive (EU) 2015/863) is the directive aimed at reducing the harmful environmental impact of waste electrical equipment by restricting the use of known dangerous substances. Based on information received from our supply sources, LD4B hereby states that the banned substances listed in the RoHS directive are not found in the parts and materials used above the threshold level listed other than exceptions approved by the European Commission.

REACH Compliance Statement

Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH) is a European Union regulation 1907/2006/EC that addresses the production and use of chemical substances, and their potential impacts on human health and the environment. Based on information received from our supply sources, LD4B hereby states compliance of the parts and materials used in manufacturing to REACH regulation. LD4B does not manufacture or import any substances or preparations as defined under REACH.